



INDIAN INSTITUTE OF TECHNOLOGY MADRAS

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Quiz I ☒Quiz II/ Mid-Sem ☐End-Semester ☐Make-up ☐

Date : 1/09/2024

Semester & Degree : 2nd, MS(R)

Course No. MA5104

Part :

Question No.	1	2	3	4	5	6	7	8	9	10
Marks	5	10	10	20	28	10				

11	12	13	14	15	16	17	18	19	20	Total
										83

Answer on both sides of the paper including the space below

1. (a) → (A) ✓

(b) → (A) ✗

(c) → (D) ✓

(d) → (A) ✗

2.

(a) False ✓

(b) False ✓

(c) False ✓

(d) True ✓

(e) False ✓

3. The error of margin is given by,

$$e = Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$(a) \hat{p} = 0.2$$

$$1-\hat{p} = 0.8$$

$$(\hat{p})(1-\hat{p}) = 0.2 \times 0.8 = 0.16$$

To find the sample size n , if $\hat{p}$ is given,

$$e = Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\frac{e}{Z_{\alpha/2}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

squaring on both sides,

$$\left(\frac{e}{Z_{\alpha/2}}\right)^2 = \frac{\hat{p}(1-\hat{p})}{n}$$

$$n = \frac{\hat{p}(1-\hat{p})(Z_{\alpha/2})^2}{e^2}$$

$$= \frac{(0.2)(0.8)(1.96)^2}{(0.05)^2}$$

$$= \frac{(0.16)(1.96)^2}{0.0025}$$

$$Z_{\alpha/2}$$

$$\alpha = 0.05$$

$$\alpha/2 = 0.025$$

$$Z_{0.025} = 1.96$$

$$e = 0.05$$

$$= \frac{0.614656}{0.0025} = 245.8624$$

rounding to higher int

$$n = 246$$

(b) if no information for p is given then, by taking the worst case scenario ($1/4$), we get,

$$n = \frac{Z_{\alpha/2}^2 \left[\frac{Y+X}{\varepsilon} \right]^2}{4c^2} \quad c = 0.05$$

$$= \frac{(1.96)^2}{4(0.05)^2} \quad Z_{0.025} = 1.96$$

$$= \frac{3.8416}{0.01} = 384.16$$

rounding to higher int

$$n = 385$$

4. X & Y are independent

$$X \sim N(20, 2)$$

$$Y \sim N(0, 7)$$

$$(1) \quad \hat{\theta}_1 = \frac{X+Y}{3}$$

$$MSE(\hat{\theta}_1) = \text{Var}(\hat{\theta}_1) + (\text{Bias}(\hat{\theta}_1))^2$$

$$\text{Bias}(\hat{\theta}_1) = E[\hat{\theta}_1] - 0$$

$$E[\hat{\theta}_1] = E\left[\frac{X+Y}{3}\right]$$

$$= \frac{1}{3} (E[X+Y])$$

$$= \frac{1}{3} (E[X] + E[Y])$$

$$E[X] = 20$$

$$E[Y] = 0$$

$$= \frac{1}{3} (20 + 0) = \frac{20}{3} = 0 = E[\hat{\theta}_1]$$

$$\text{Bias}[\hat{\theta}_1] = 0 - 0 = 0$$

Therefore, unbiased

$$\text{Var}(\hat{\theta}_1) = \text{Var}\left(\frac{X+Y}{3}\right) = \frac{1}{9} (\text{Var}(X+Y))$$

$$= \frac{1}{9} (\text{Var}(X) + \text{Var}(Y))$$

$$\text{Var}(X) = 2$$

$$\text{Var}(Y) = 7$$

$$= \frac{1}{9} (2+7) = \frac{9}{9} = 1$$

since, $\text{bias}(\hat{\theta}_1) = 0$

$$\text{MSE}(\hat{\theta}_1) = \text{Var}(\hat{\theta}_1) = \boxed{1.1}$$

$$(2) \hat{\theta}_2 = \frac{4X - Y}{7}$$

$$E[\hat{\theta}_2] = E\left[\frac{4X - Y}{7}\right]$$

$$= \frac{1}{7} E[4X - Y] = \frac{1}{7} [E[4X] - E[Y]]$$

$$= \frac{1}{7} (4E[X] - E[Y]) \quad \begin{matrix} E[X] = 20 \\ E[Y] = 0 \end{matrix}$$

$$= \frac{1}{7} (80 - 0) = \frac{80}{7} = 0$$

$$\text{Bias}(\hat{\theta}_2) = E[\hat{\theta}_2] - 0$$

$$= 0 - 0 = 0 \quad \therefore \text{unbiased}$$

$$\text{Var}(\hat{\theta}_2) = \text{Var}\left(\frac{4X - Y}{7}\right) = \frac{1}{49} \text{Var}(4X - Y)$$

$$= \frac{1}{49} [16 \text{Var}(X) - \text{Var}(Y)]$$

$$= \frac{1}{49} (32 - 7) = \frac{25}{49}$$

$$\text{Var}(X) = 2$$

$$\text{Var}(Y) = 7$$

$$\text{Var}(\hat{\theta}_2) = 0.51020$$

since $\text{bias}(\hat{\theta}_2) = 0$

$$\therefore \text{MSE}[\hat{\theta}_2] = \text{Var}(\hat{\theta}_2) = \boxed{0.51020}$$

(3) Since both estimators $\hat{\theta}_1$ & $\hat{\theta}_2$ has both bias as 0. But, we will prefer $\hat{\theta}_2$ since, the variance is lower for $\hat{\theta}_2$ than $\hat{\theta}_1$.

When we have 2 estimators with same bias, we choose the estimator with lower variance as it provides lesser variability.

5.

$$n_1 = 14, \quad \bar{X}_1 = 17, \quad S_1^2 = 1.5$$

$$n_2 = 16, \quad \bar{X}_2 = 19, \quad S_2^2 = 1.8$$

(a) CI = 99%

$$\alpha = 0.01, \quad \alpha/2 = 0.005$$

normal populaⁿ with equal variances, therefore we pool the sample variances

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(13)(1.5) + (15)(1.8)}{28}$$

$$= \frac{19.5 + 27}{28} = 1.660714 \quad /$$

$$S_p = \sqrt{S_p^2} = 1.288687$$

CI for $\mu_1 - \mu_2$,

$$= \bar{X}_1 - \bar{X}_2 \pm t_{\alpha/2, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

$$= -2 \pm (2.763)(1.28867)(0.36596) \quad t_{0.005, 28} = 2.763$$

$$= -2 \pm 1.30303$$

$$= \boxed{(-3.30303, -0.69697)}$$

(10)

$$\sqrt{\frac{1}{14} + \frac{1}{16}}$$

$$\sqrt{\frac{16+14}{224}}$$

$$\sqrt{\frac{30}{224}}$$

$$= 0.36596$$

(b) since, $\mu_1 - \mu_2 < 0$

our entire interval also lies than 0.

Then that means, we can chose medicine 1 as it already has the lower mean. The goal is to make sure to reduce the length of time to recovery

interpretation)

(3)

(c) CI for $\frac{\sigma_1^2}{\sigma_2^2}$

$$\frac{s_1^2}{s_2^2} \frac{1}{F_{\alpha/2}(v_1, v_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} F_{\alpha/2}(v_2, v_1)$$

$$v_1 = 14 - 1 = 13$$

$$v_2 = 16 - 1 = 15$$

$$\alpha = 0.1$$

$$\alpha/2 = 0.05$$

$$\frac{s_1^2}{s_2^2} = \frac{1.5}{1.8} = 0.8333$$

$$F_{0.05}(13, 15) = 2.45$$

$$F_{0.05}(15, 13) = 2.53$$

$$\frac{0.8333}{2.45} < \frac{\sigma_1^2}{\sigma_2^2} < (0.8333)(2.53)$$

$$\boxed{0.34012 < \frac{\sigma_1^2}{\sigma_2^2} < 2.108249} \quad 10$$

(d) Since $\frac{\sigma_1^2}{\sigma_2^2} = 1 \Rightarrow \sigma_1^2 = \sigma_2^2$

and $1 \in [0.34012, 2.108249]$

It was fine ~~that~~ I justified that we assumed $\sigma_1^2 = \sigma_2^2$ earlier while creating the confidence interval for $\mu_1 - \mu_2$

6. $X_1, \dots, X_n \sim \text{Poisson}$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$x = 0, 1, 2, \dots$$

$$\lambda > 0$$

(a) given $E(X) = \lambda$

then we can say,

$$\bar{X} = \lambda$$

then $\boxed{\hat{\lambda}_{\text{MOM}} = \bar{X}}$

(b) $P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$

$$L(\lambda) = \prod \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$$

$$= e^{-n\lambda} \lambda^{\sum x_i} \frac{1}{\prod (x_i!)}$$

$l(\lambda) = \log \text{likelihood of } L(\lambda)$

first derivative $= -n\lambda + \sum x_i \log \lambda + c$

$$l'(\lambda) = -n + \frac{\sum x_i}{\lambda}$$

equate it to 0,

$$-n = -\frac{\sum x_i}{\lambda}$$

this term will be anyways dropped

second derivative?

$$\boxed{\hat{\lambda}_{\text{MLE}} = \bar{X}}$$

$$\lambda = \frac{\sum x_i}{n} = \bar{X}$$

(10)